

@porun8-roi

INORGANIC CHEMISTRY

JEE (MAIN + ADVANCED)

XIII

IMPORTANT NOTES ON QUALITATIVE SALT ANALYSIS

by PMS

Dry Test :

Dry tests are preliminary test which are used for detection of s-block cations and coloured d-block paramagnetic cations, these tests are of following types.

- (i) Flame test (Usually applied for detection of s-block cations)
 - (ii) Borax Bead Test
 - (iii) Phosphate Bead Test
- } Applied for coloured d-block paramagnetic cations)
- (iv) Cobalt Nitrate Test (Applied for few specific cations)

(1) Flame Test :

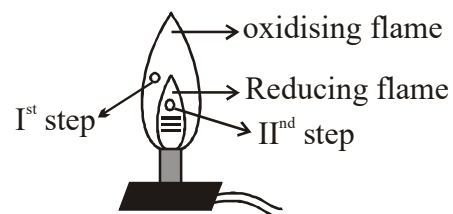
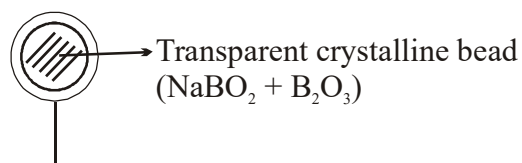
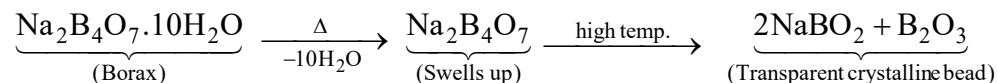
It is performed with the help of platinum coated wire which is first dipped in conc. HCl and brought in contact with given solid mixture. In this way, given salts are converted into more volatile chloride salt which impart characteristic colour to flame except Be^{2+} and Mg^{2+} cation (do not impart colour due to high ionization energy)

Cations	Colour of Flame
Li^+ Na^+ K^+ Rb^+ Cs^+	Carmin Red Golden Yellow Lilac. (Pale violet) Reddish violet Blue
Ca^{2+} Sr^{2+} Ba^{2+}	Brick Red Crimson Red Apple Green
Cu^{2+} $\text{Pb}^{2+}, \text{Bi}^{3+}$	Greenish blue Blue

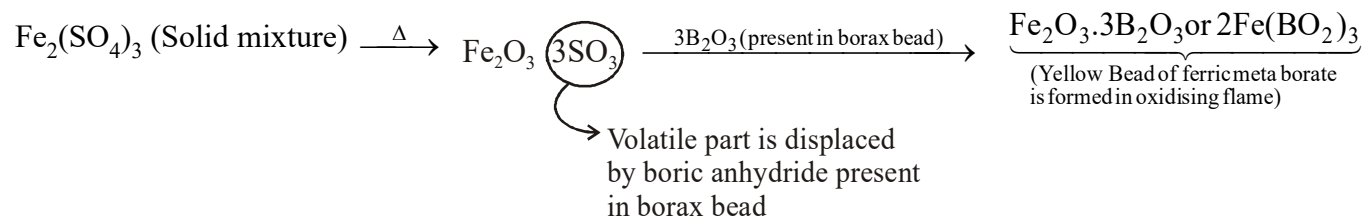


(2) Borax Bead Test :

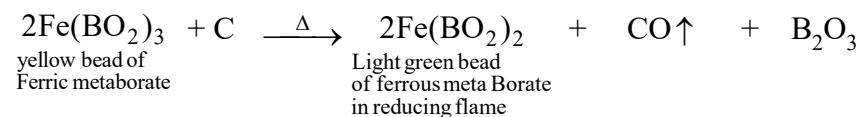
Formation of Bead : When Borax is heated on platinum coated wire loop, then it first swells up due to loss of water of crystallisation and covers entire portion of loop, which on further heating gives transparent crystalline Bead of sodium meta borate and Boric Anhydride ($\text{NaBO}_2 + \text{B}_2\text{O}_3$).



Reaction in Oxidising Flame : (Ist Step)



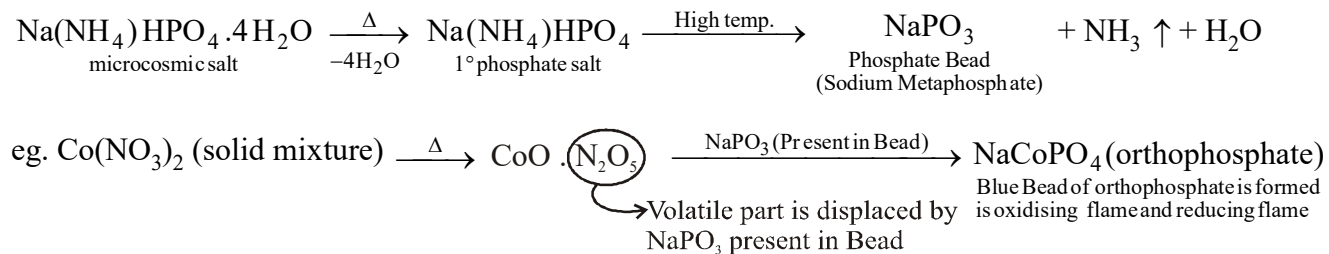
Reaction in Reducing Flame : (IInd Step)



Coloured d-block cations	Bead colour in oxidising flame	Bead colour in reducing flame	Product transformation
Cu^{2+}	Blue	Red	$\text{Cu}(\text{BO}_2)_2 \rightarrow \text{Cu}$
Fe^{3+}	Yellow	Green	$\text{Fe}(\text{BO}_2)_3 \rightarrow \text{Fe}(\text{BO}_2)_2$
Cr^{3+}	Green	Green	$\text{Cr}(\text{BO}_2)_3 \rightarrow \text{Cr}(\text{BO}_2)_3$
Co^{2+}			$\text{Co}(\text{BO}_2)_2 \rightarrow \text{Co}(\text{BO}_2)_2$
Ni^{2+}	Brown	Grey	$\text{Ni}(\text{BO}_2)_2 \rightarrow \text{Ni}$
Mn^{2+}	Amethyst(violet)	Colourless	$\text{Mn}^{+3}(\text{BO}_2)_3 \rightarrow \text{Mn}^{+2}(\text{BO}_2)_2$

Note : All above colour of bead are observed in their cold condition (without touching flame.)

(3) Phosphate Bead Test :

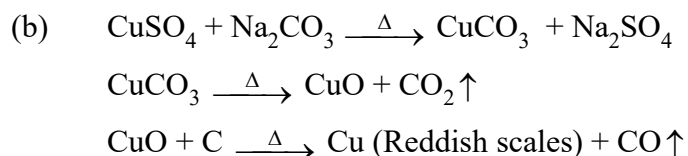
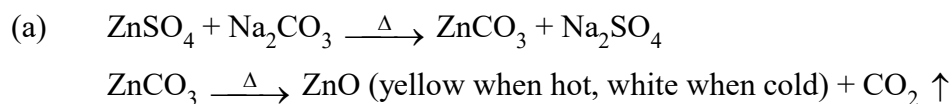


Note : Colour table of phosphate Bead test is same as Borax Bead test in cold condition (without touching the flame).

(4) Charcoal Cavity Test :

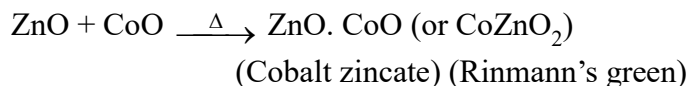
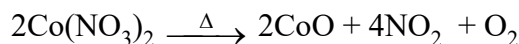
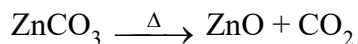
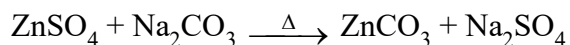
This test is based on the fact that metallic carbonates when heated in a charcoal cavity decompose to give corresponding oxides. The oxides appear as coloured residue in the cavity. In certain cases, the oxides formed partially undergo reduction to the metallic state producing metallic scales.

Example



Cobalt Nitrate Charcoal Cavity Test :

In case the residue is white during cavity test then one or two drops of cobalt nitrate solution is added and again heated in the oxidizing flame using blow pipe, different metal salts give different coloured mass as given in the table.

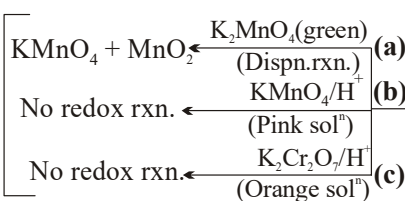


S.No.	Metal	Colour of the mass
1.	Zinc	CoO.ZnO Rinmann 's green
2.	Alu min ium	CoO.Al ₂ O ₃ Thinard blue

Carbonate (CO_3^{2-}) / Bicarbonate (HCO_3^-)

1. All CO_3^{2-} salts are water insoluble except alkali metal carbonate salts and $(\text{NH}_4)_2\text{CO}_3$.
2. All metal bicarbonate salts are water soluble. Except NaHCO_3 (Sparingly soluble)

H_2CO_3 is non-oxidising and hence $\text{CO}_3^{2-}/\text{CO}_2$ does not act as oxidant



(4) Test based on redox prop.

Observation [Colourless odourless gas evolves with brisk effervescence]

CO_2 gas

Identification of liberated gas (Lab test)

(2)

Warm (1) dil. HCl/ H_2SO_4

Distinction between CO_3^{2-} & HCO_3^-

(5)

Excess BaCl_2 solⁿ & filter

(a)

R.T.
 $\text{BaCO}_3 \downarrow$
white ppt.

(b)

If filtrate contains HCO_3^- ion

Boil
 $\text{BaCO}_3 \downarrow$
white ppt.

Base (NH_3 solⁿ few drops)
 $\text{BaCO}_3 \downarrow$
white ppt.

(Salt solⁿ)

$\text{CO}_3^{2-}(\text{aq.})/\text{HCO}_3^-(\text{aq.})$

(3) Precipitation test

(a) BaCl_2 solⁿ

$\text{BaCO}_3 \downarrow$
white
 $\downarrow$ dil. HCl
 BaCl_2 (solⁿ)

(b) $\text{Pb}(\text{OAc})_2$ sol.

$\text{PbCO}_3 \downarrow$
white ppt.
 $\downarrow$ dil. HNO_3
 $\text{Pb}(\text{NO}_3)_2$ (solⁿ)

(c) AgNO_3 solⁿ

$\text{Ag}_2\text{CO}_3 \downarrow$
white/yellow ppt.

excess NH_3 solⁿ
 $[\text{Ag}(\text{NH}_3)_2]^+$ (soluble)

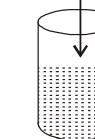
Boil
 $\text{Ag}_2\text{O} \downarrow$
Brown ppt.
 $\updownarrow$ excess NH_3 solⁿ

$1/2\text{N}_2 \uparrow + 3\text{Ag} \xleftarrow[\text{I.M.R.}]{\text{explosion}}$

Ag_3N
Silver nitride
(Fulminating silver)

On standing $[\text{Ag}(\text{NH}_3)_2]^+ (\text{aq})$

(a)



$\text{Ca}(\text{OH})_2$
(Lime water)

$\downarrow \text{CO}_2$

CaCO_3
(white turbidity)

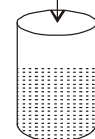
excess CO_2

$\text{Ca}(\text{HCO}_3)_2$
(colourless solⁿ)

Boil

CaCO_3
(white turbidity)

(b)



$\text{Ba}(\text{OH})_2$
(Baryta water)

$\downarrow \text{CO}_2$

BaCO_3
(white turbidity)

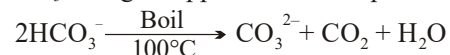
excess CO_2

$\text{Ba}(\text{HCO}_3)_2$
(colourless solⁿ)

Boil

BaCO_3
(white turbidity)

Note : 1. CO_3^{2-} ion gives ppt. at room temp. whereas HCO_3^- ion forms ppt. of carbonate on boiling.



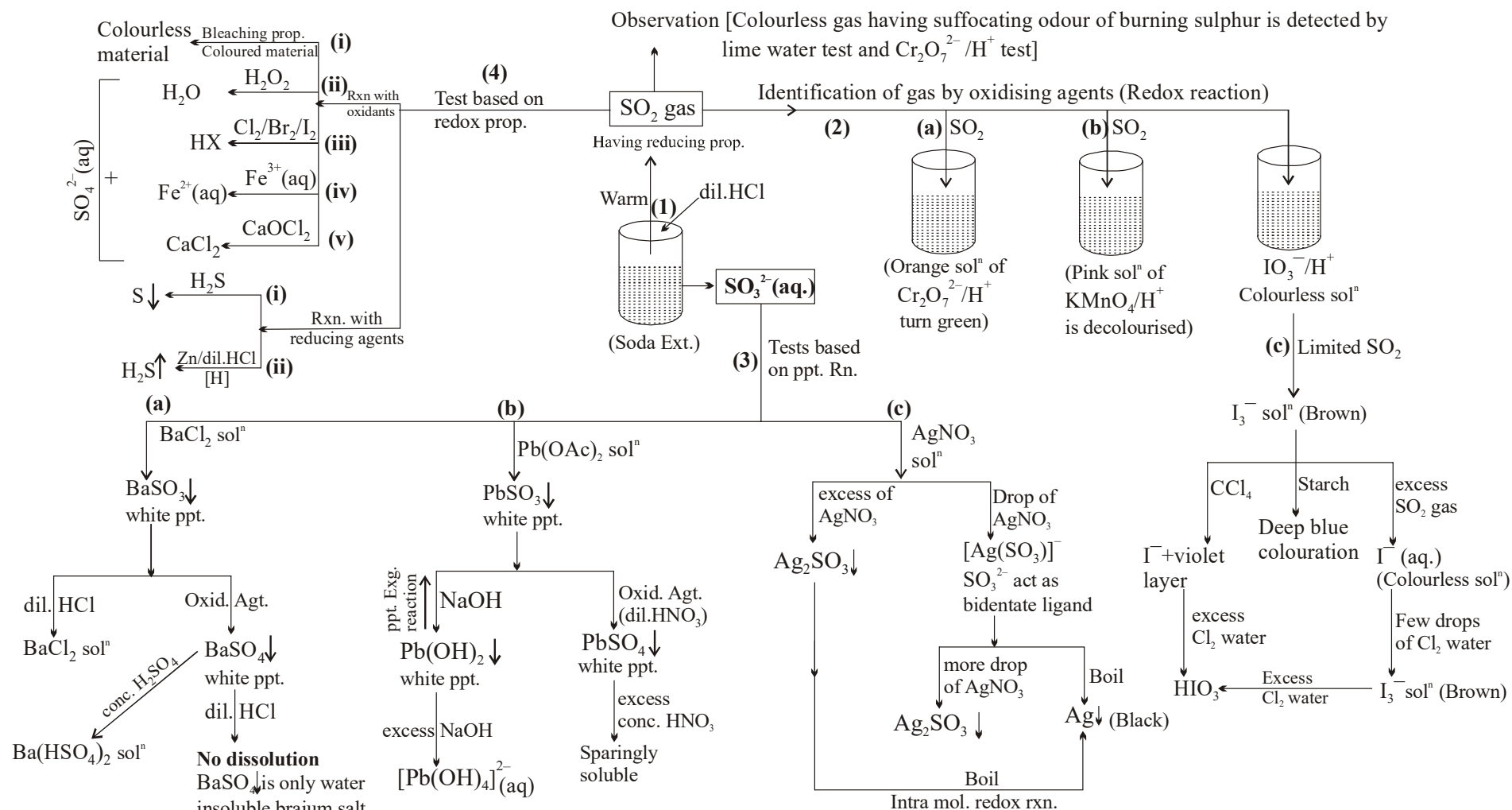
2. Water insoluble carbonate salts are soluble in soda water ($\text{CO}_2 + \text{H}_2\text{O}$) and CH_3COOH .

I.M.R. = Inter Molecular Redox



Sulphite (SO_3^{2-})

1. All SO_3^{2-} salt are water insoluble except alkali metal sulphite salts and $(\text{NH}_4)_2\text{SO}_3$



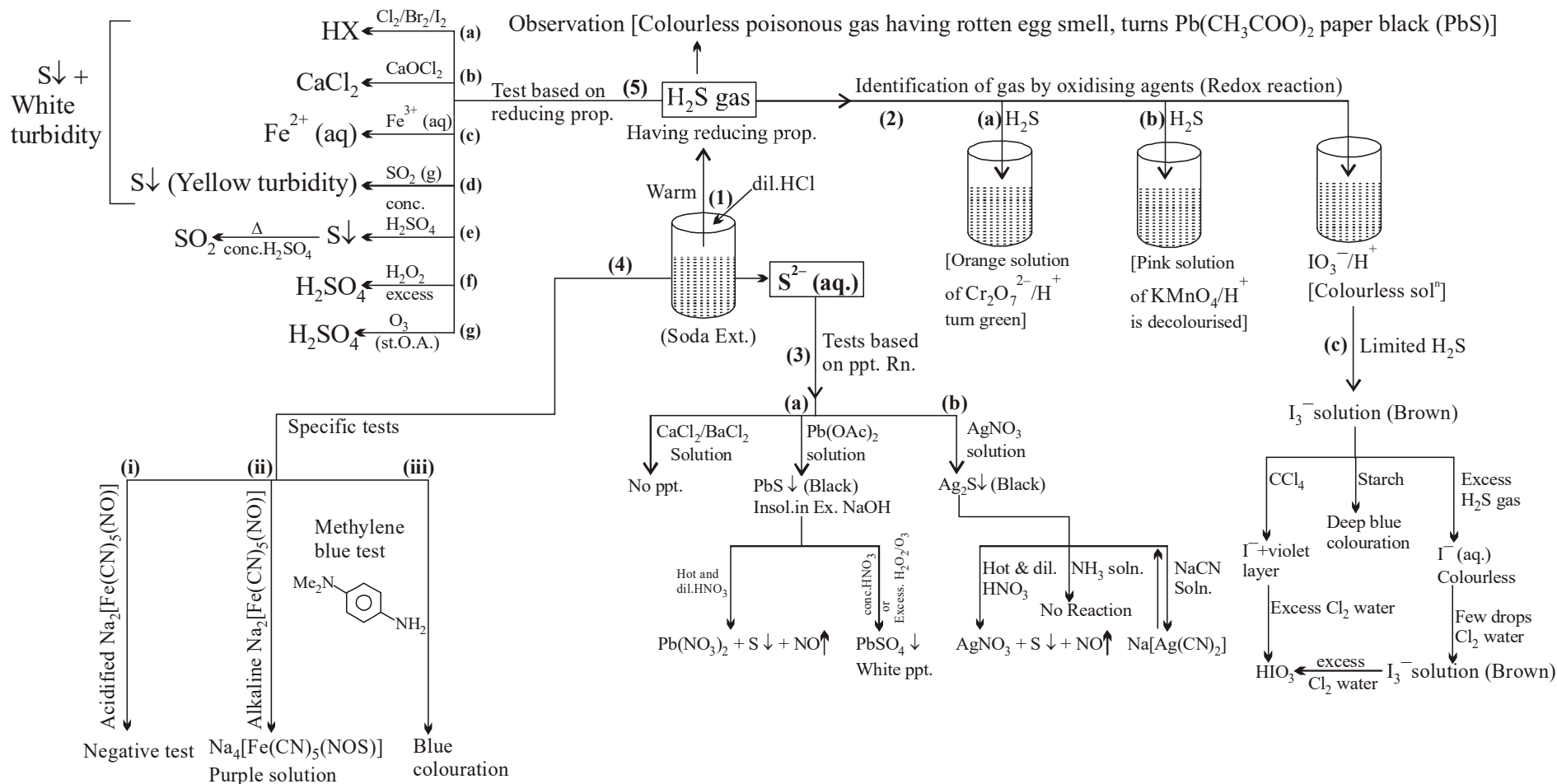
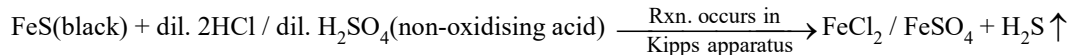
Note : 1. As SO_2 gas is more soluble in water, therefore effervescence is not observed during evolution of SO_2 gas.

2. Bleaching action due to oxidising property is permanent, whereas bleaching action due to reducing prop. is temporary.

3. $\text{H}_2\text{SO}_3/\text{SO}_2$ exhibit better reducing properties and weaker oxidising property.

@poru.in8-Sulphide (S²⁻)

- All S²⁻ salts of s-block metals are water soluble and colourless whereas S²⁻ salts of p-block & d-block are water insoluble and usually coloured due to polarisation effect.
- S²⁻ salts of Al³⁺, Cr³⁺ and Mg²⁺ do not exist in aqueous solution because they readily hydrolysed into corresponding metal hydroxide due to their low K_{sp}, hence they are prepared in dry condition. Fe³⁺ shows redox reaction with S²⁻ ions. [Fe³⁺ + S²⁻ → Fe²⁺ + S↓]
- FeS↓ is the only black metal sulphide salt decomposed by non-oxidising acid (like dil. HCl/dil. H₂SO₄) whereas other remaining black sulphide are dissolved in oxidising acid (hot and dil. HNO₃) except HgS↓ (Insoluble in hot and dil. HNO₃ but soluble in aquaregia and S²⁻ salt solution)

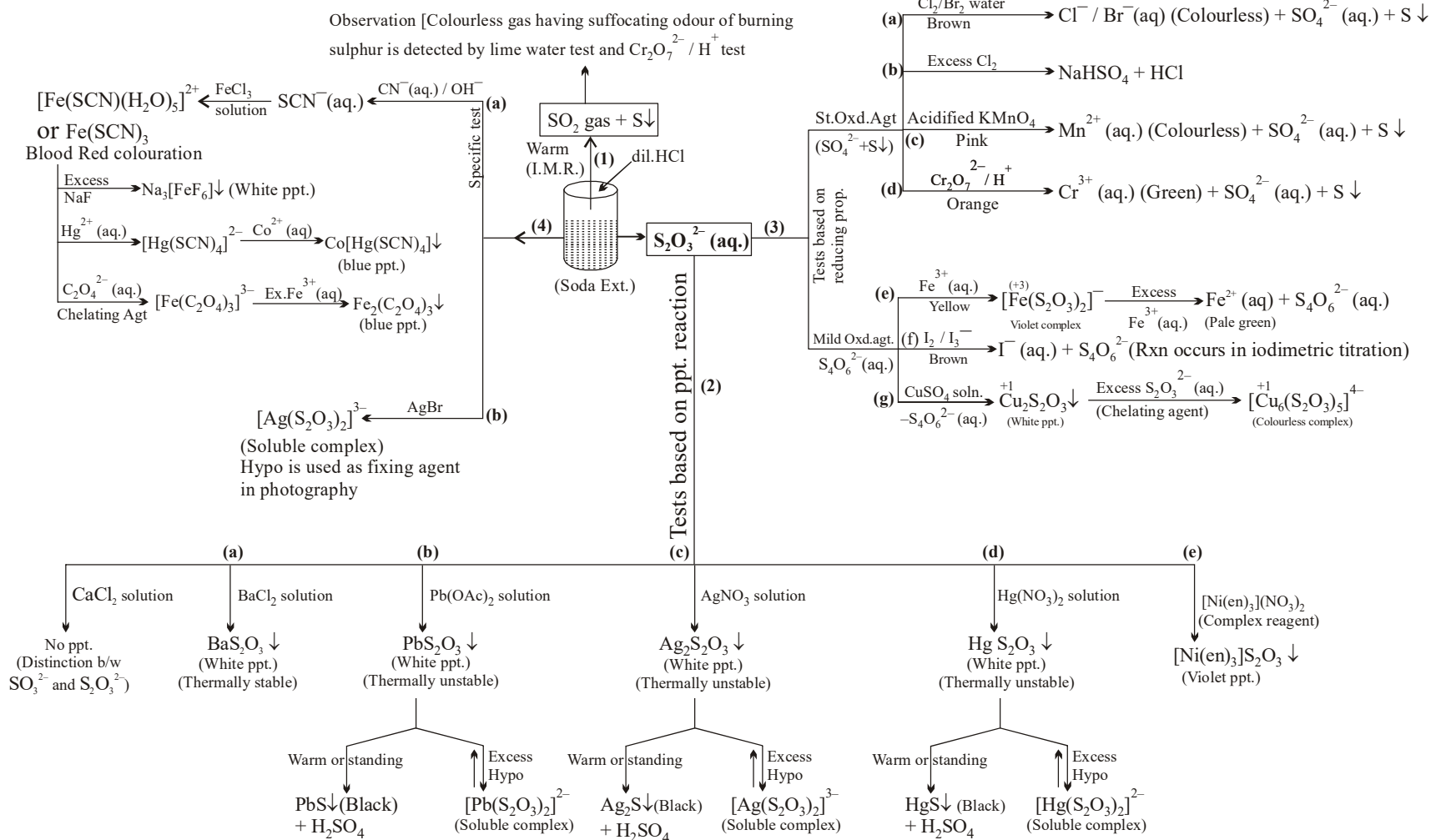


- Note :**
- Mⁿ⁺ (p and d-block metal cation) forms metal sulphides with H₂S gas depending upon acidic, neutral, basic medium of solution
 - PbS↓ (black) is insoluble in excess NaOH solution or excess NH₄OH solution due to its low K_{sp}
 - Specific test with Na₂[Fe(CN)₅NO] (sodium nitroprusside) requires high conc. of S²⁻ ion, hence neutral H₂S and acidified H₂S gives negative test, whereas alkaline H₂S gives purple colouration.



Thiosulphate ($\text{S}_2\text{O}_3^{2-}$)

1. All $\text{S}_2\text{O}_3^{2-}$ salts of s-block are water soluble except : $\text{BaS}_2\text{O}_3 \downarrow$ (white ppt.)
2. $\text{S}_2\text{O}_3^{2-}$ salts of Ag^+ , Bi^{3+} , Cu^+ , $[\text{Ni(en)}_3]^{2+}$, Hg_2^{2+} , Hg^{2+} , Ba^{2+} , and Pb^{2+} are water insoluble.

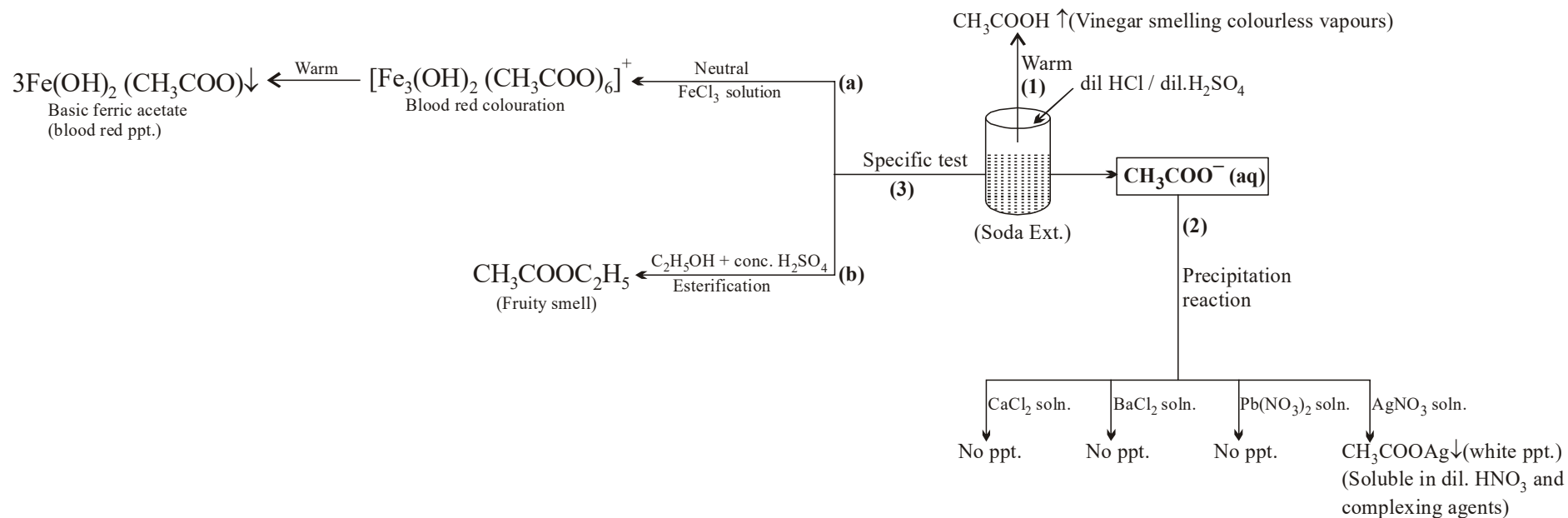


- Note :**
1. $\text{Na}_2\text{S}_2\text{O}_3$ (Hypo. soln) mainly exhibits reducing character.
 2. Hypo solution acts as Antichlore (removes excess Cl_2 from bleached cotton fabric) like SO_2 .



Acetate (CH_3COO^-)

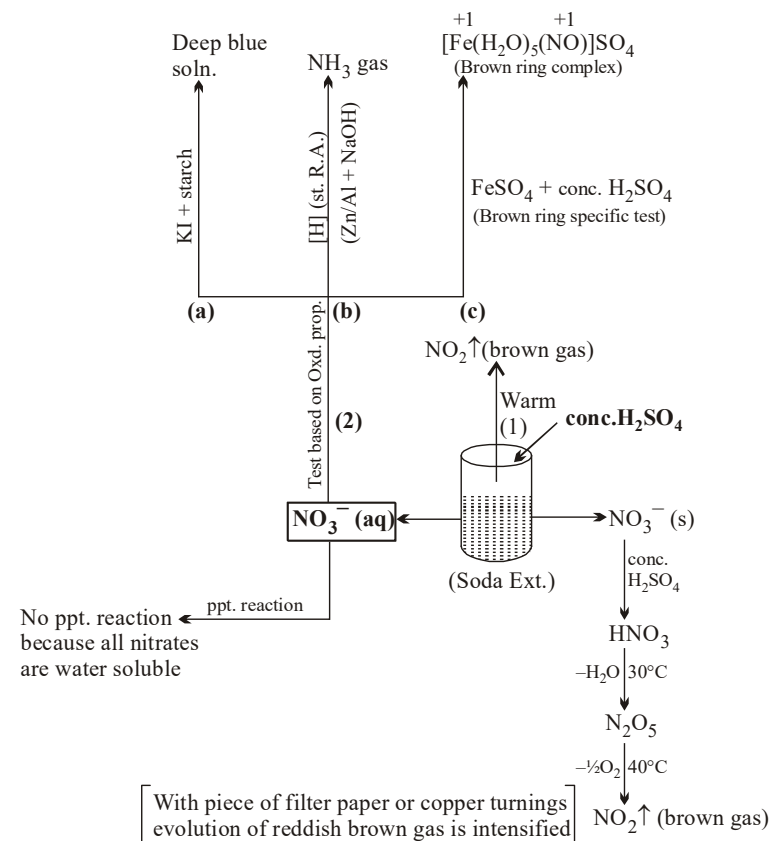
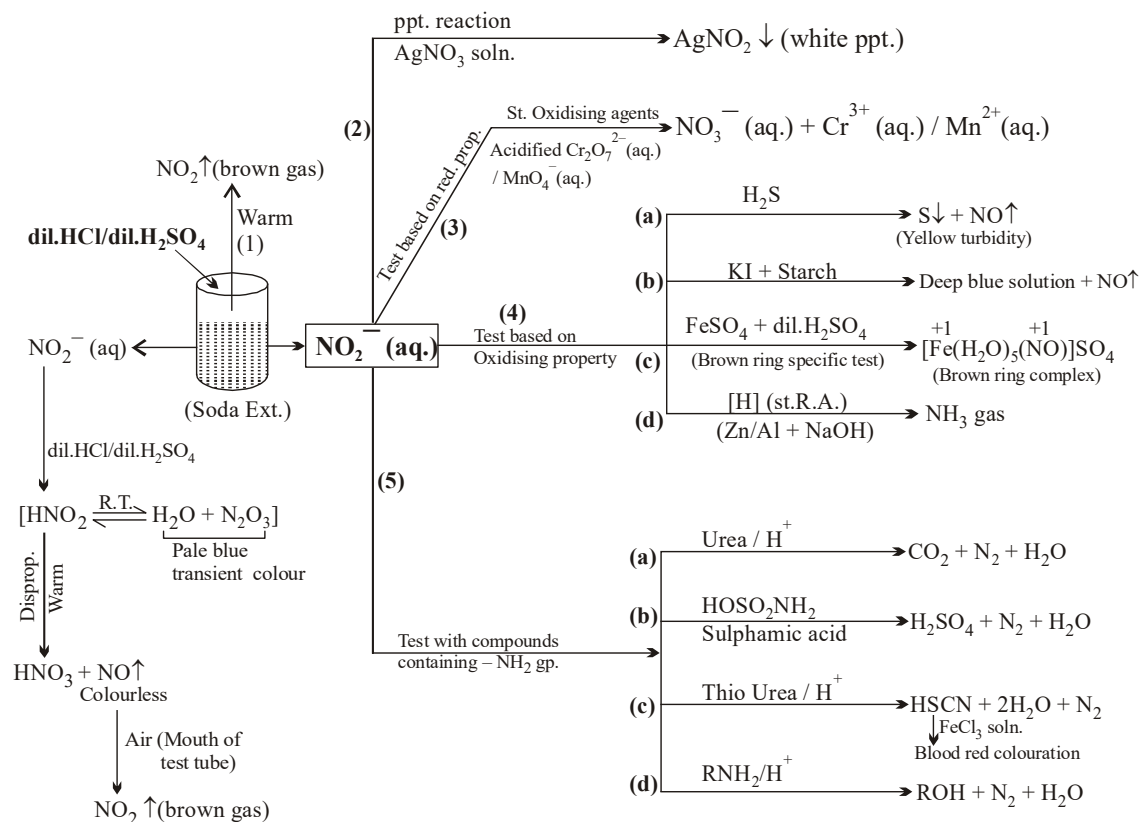
1. All CH_3COO^- salts are water soluble except CH_3COO^- salt of Ag^+ , Hg_2^{2+} , Cu^+



Note : Deep red colouration is obtained by reaction of Fe^{3+} cation with SCN^- (aq) and CH_3COO^- (aq).

Nitrite (NO_2^-) and Nitrate (NO_3^-)

1. All NO_2^- salts are water soluble except NO_2^- salt of Ag^+ , Hg_2^{2+} , Cu^+
2. All NO_3^- salts are water soluble.

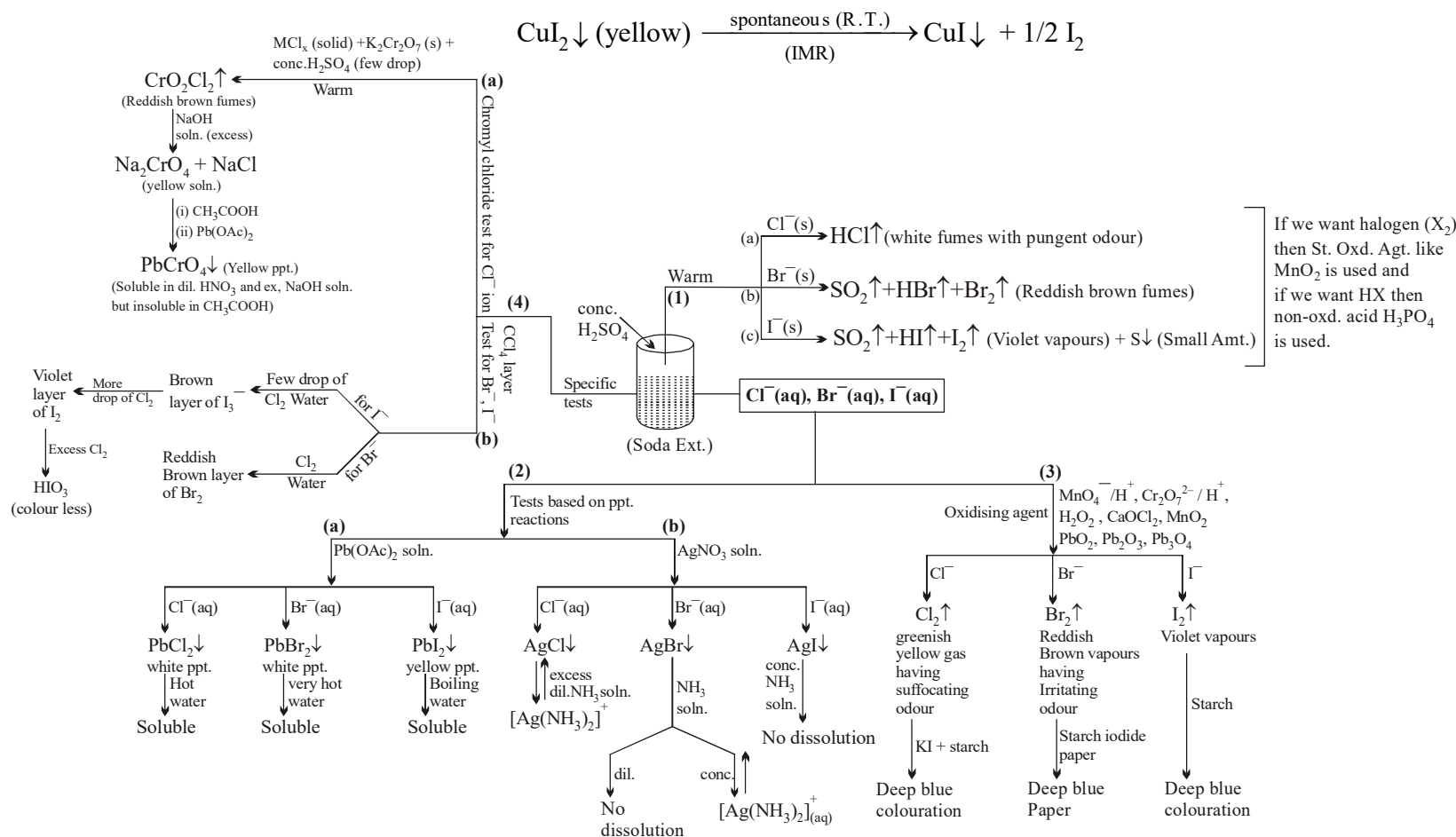


- Note:**
1. Aq. soln. of $\text{M}(\text{NO}_2)_2$ / $\text{M}(\text{NO}_3)_2$ salts ($\text{M}^{2+} = \text{Ba}^{+2}$, Sr^{+2} , Pb^{2+}) do not give brown ring test because these cations form insoluble SO_4^{2-} salts with H_2SO_4 used in ring test, formation of ppt. will create turbulence in soln. therefore, brown ring complex is not formed. In such cases, above test can be modified by first adding Na_2SO_4 soln. Now obtained filtrate can be used for ring test.
 2. Ring test for NO_3^- ion in presence of NO_2^- (aq) should not be done however ring test for NO_2^- (aq) ion in presence of NO_3^- (aq) can be performed.
 3. HNO_2 exhibits better oxidising character and weak reducing character.



Chloride (Cl^-), Bromide (Br^-) and Iodide (I^-)

1. All Cl^- salts are water soluble except $\text{AgCl} \downarrow$, $\text{Hg}_2\text{Cl}_2 \downarrow$, $\text{PbCl}_2 \downarrow$, $\text{CuCl} \downarrow$ (all are white ppt.)
2. All Br^- salts are water soluble except $\text{AgBr} \downarrow$ (pale yellow) $\text{PbBr}_2 \downarrow$ (white), $\text{Hg}_2\text{Br}_2 \downarrow$ (yellow), $\text{CuBr} \downarrow$ (white)
3. All I^- salts are water soluble except $\text{AgI} \downarrow$ (yellow), PbI_2 (yellow), $\text{HgI}_2 \downarrow$ (scarlet red), Hg_2I_2 (green), $\text{BiI}_3 \downarrow$ (black), $\text{CuI} \downarrow$ (white)



Note : 1. $\text{AgCl} \downarrow$, $\text{PbCl}_2 \downarrow$, $\text{Hg}_2\text{Cl}_2 \downarrow$, $\text{CuCl} \downarrow$, HgCl_2 (pred. covt.) PbCl_4 (pred. covt.) SnCl_4 (pred. covt.) do not give chromyl chloride test due to low concentration of Cl^- ion.

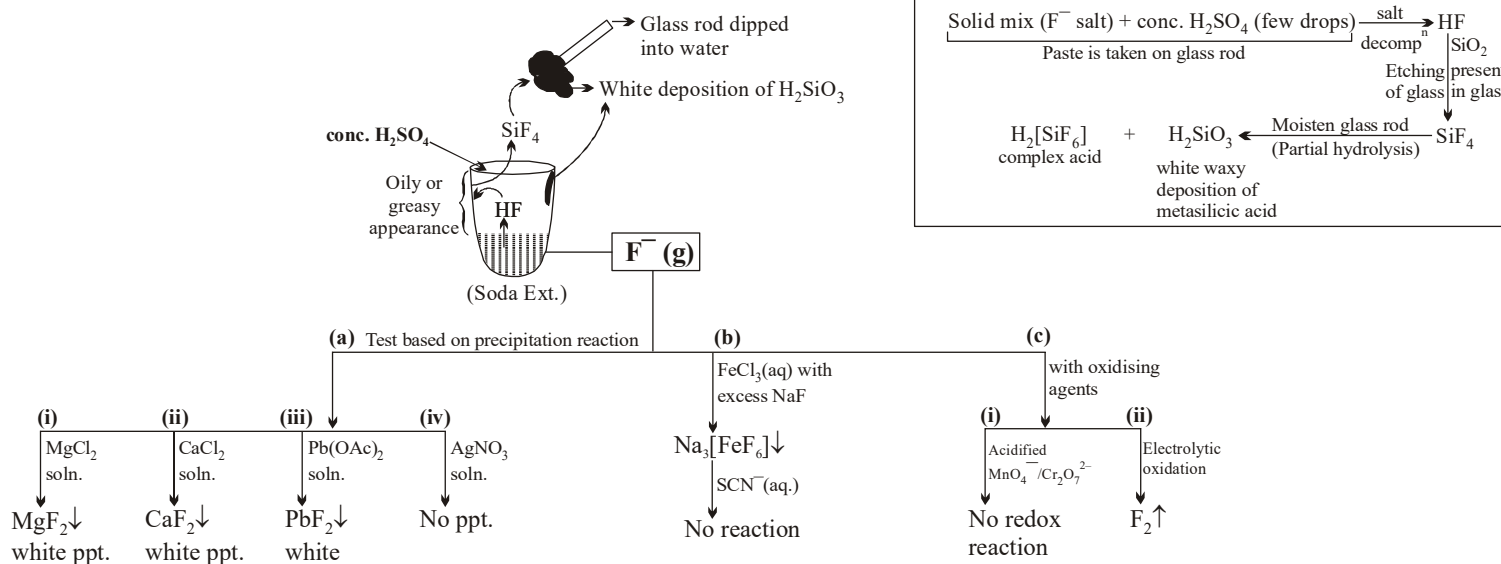
2. CH_3COOH used in chromyl chloride test neutralises excess NaOH otherwise yellow ppt. of $\text{PbCrO}_4 \downarrow$ is not formed.



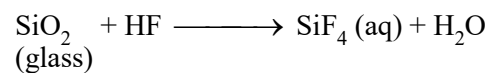
Fluoride (F⁻)

1. MgF₂, CaF₂, SrF₂ and PbF₂ are water insoluble fluoride

(1) Specific Test for F⁻ ion.



Note : HF is only acid which reacts with glass (Etching of glass) therefore in Lab it is kept in wax coated glass vessel.



Oxylate ($\text{C}_2\text{O}_4^{2-}$)

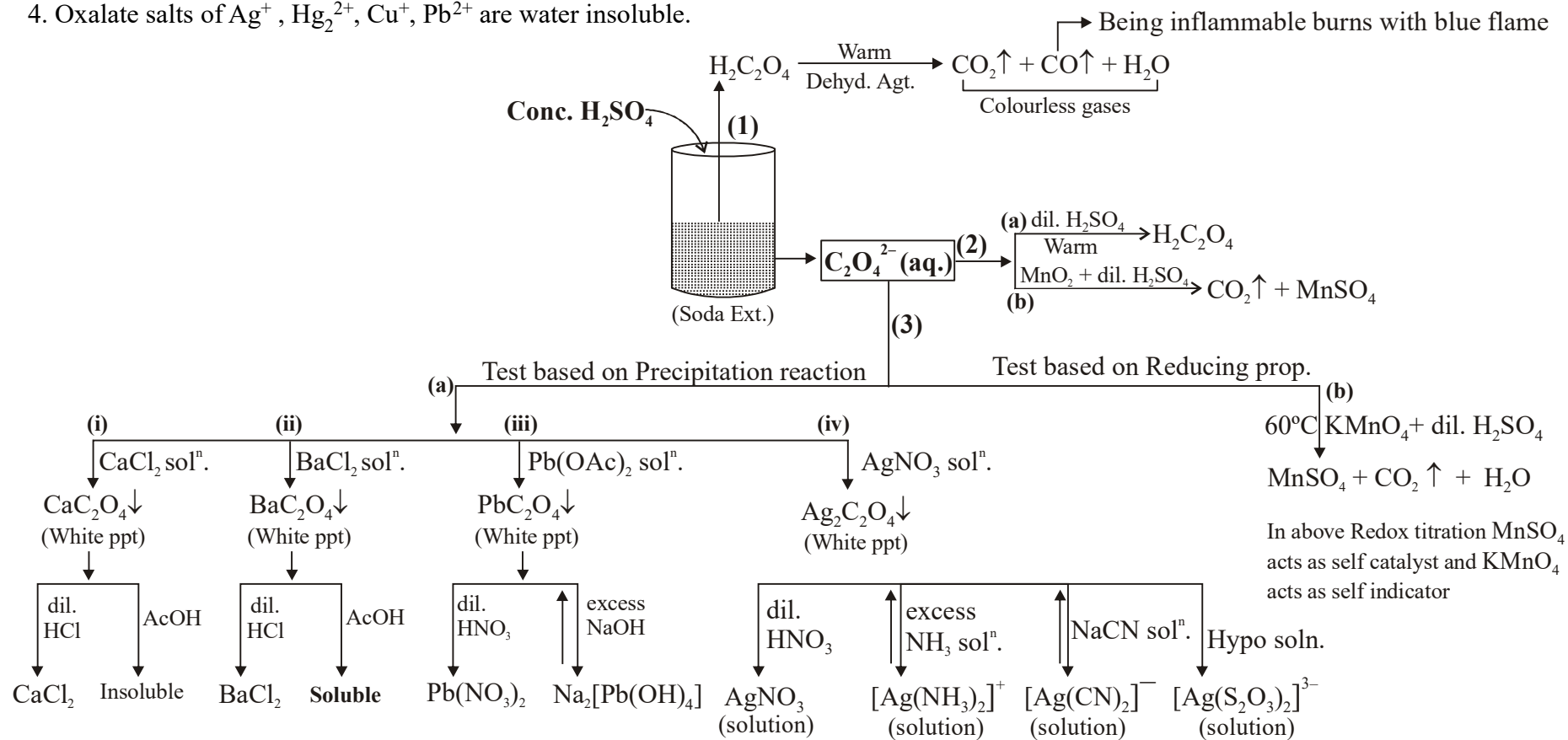
1. Oxalate salt of alkali metals are water soluble.
2. Oxalate salts of alkaline earth metals are either water insoluble or sparingly water soluble except BeC_2O_4 (completely water soluble)

order of solubility in water : $\text{MgC}_2\text{O}_4 \downarrow < \text{CaC}_2\text{O}_4 \downarrow < \text{SrC}_2\text{O}_4 \downarrow < \text{BaC}_2\text{O}_4 < \text{BeC}_2\text{O}_4$ (order is similar to F^- salt)

Water insoluble water soluble

3. $\text{BaC}_2\text{O}_4 \downarrow$ is only water insoluble oxalate salt dissolved in weak acetic acid due to its relative high K_{sp} at R.T.

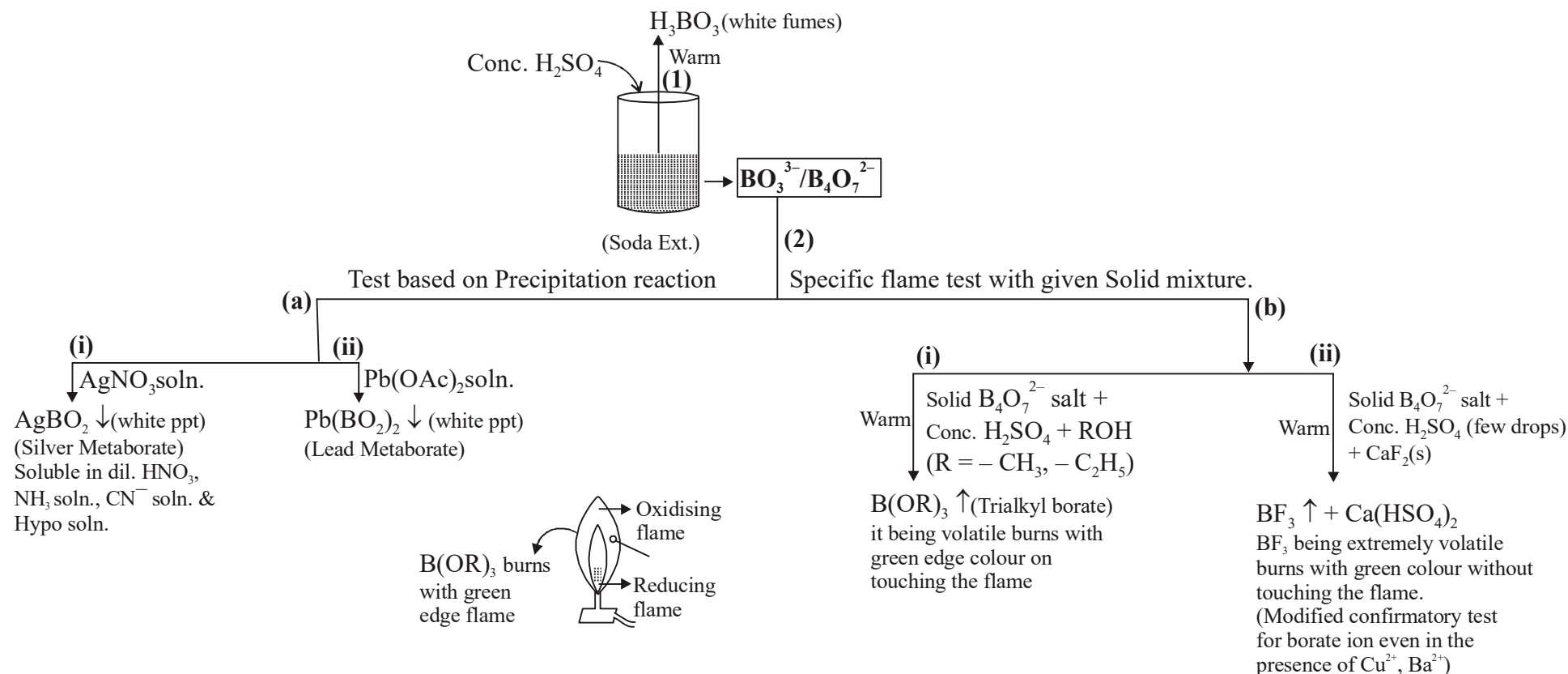
4. Oxalate salts of Ag^+ , Hg_2^{2+} , Cu^+ , Pb^{2+} are water insoluble.



Note : $\text{H}_2\text{C}_2\text{O}_4 / \text{C}_2\text{O}_4^{2-}$ can act as reductant but does not act as oxidant.

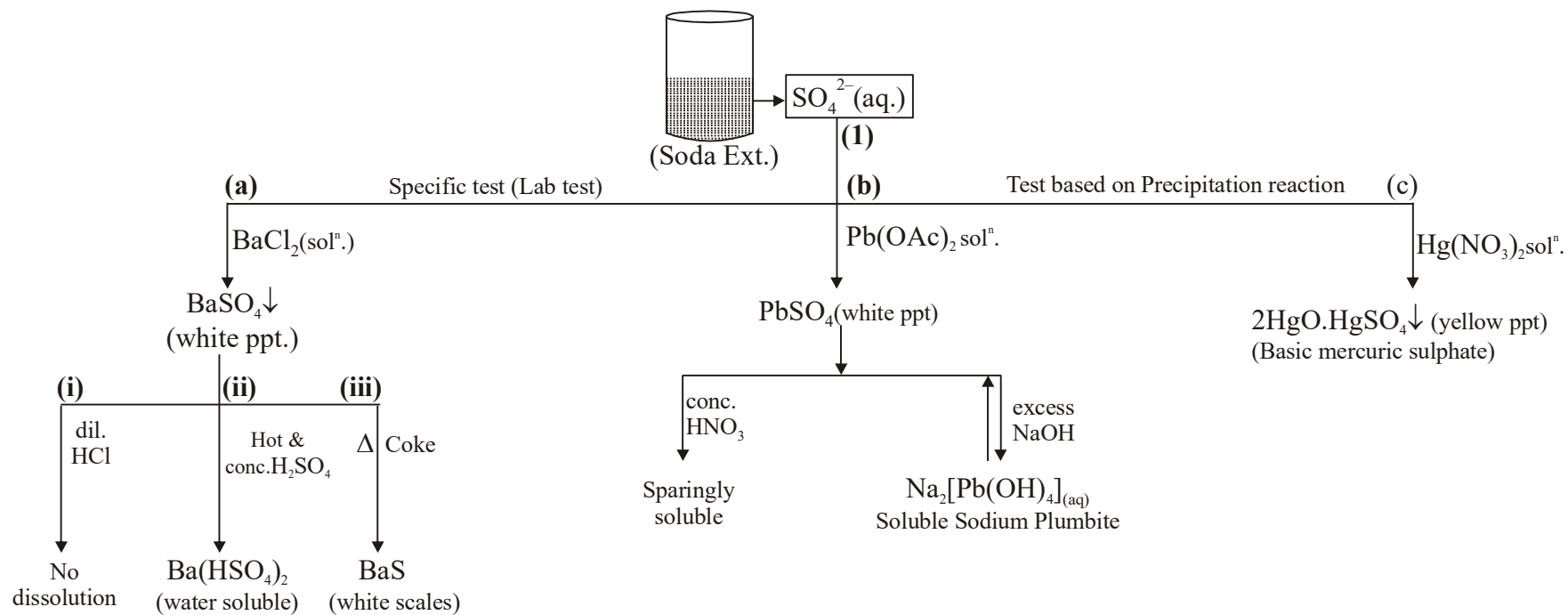
Borate (BO_3^{3-} / $\text{B}_4\text{O}_7^{2-}$)

1. Borate salts of s-block are water soluble and other borate salts are either water soluble or sparingly water soluble.
2. Very few borate salts derived from boric acid (H_3BO_3) are known, mostly borate salts are derived either from metaboric acid (HBO_2) or tetraboric acid ($\text{H}_2\text{B}_4\text{O}_7$)



SO_4^{2-} (aq) Anion

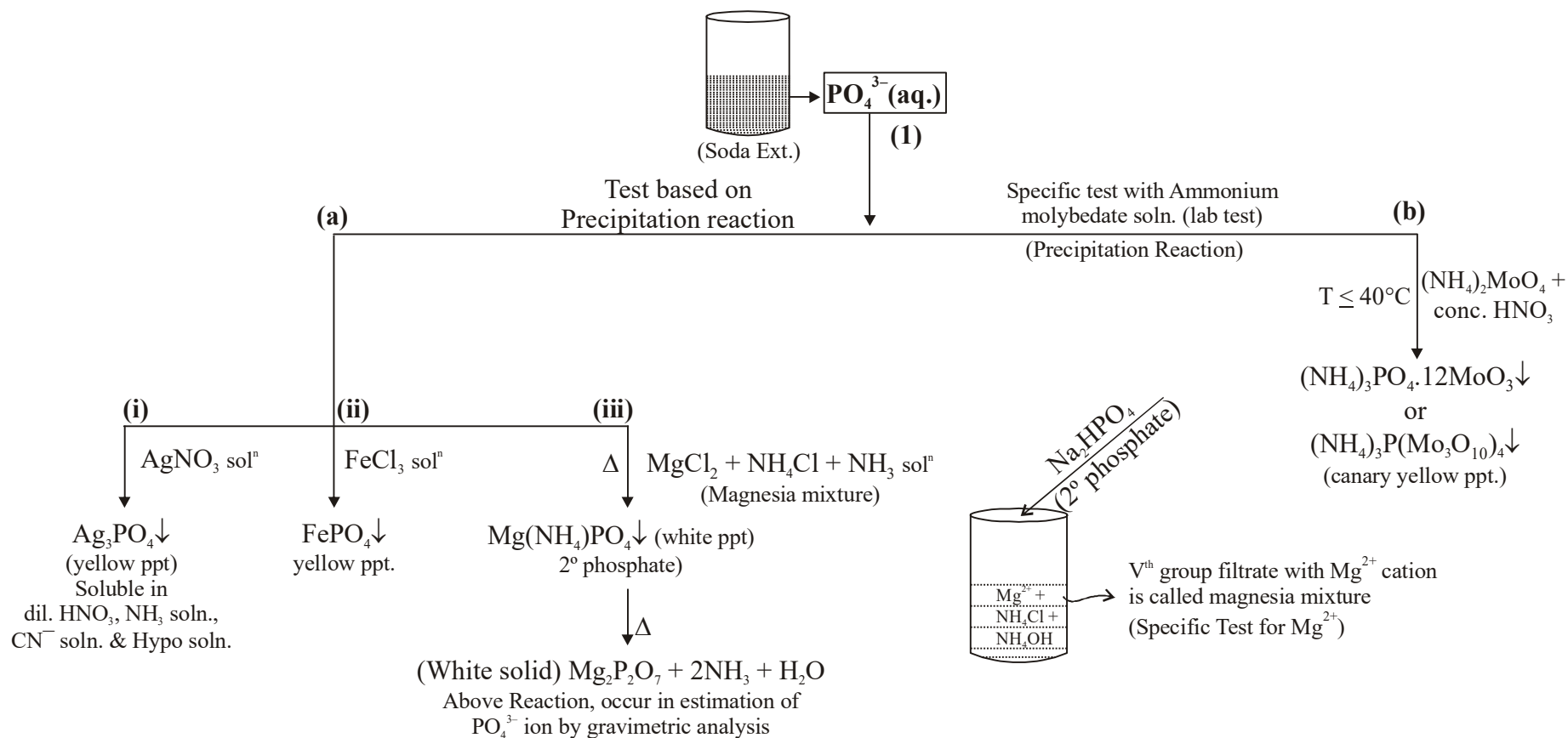
All SO_4^{2-} salts are water soluble except sulphate salts of Ba^{2+} , Sr^{2+} , Pb^{2+} and Hg^{2+}



Note : $\text{BaSO}_4\downarrow$, $\text{SrSO}_4\downarrow$ and $\text{PbSO}_4\downarrow$ salts are insoluble in any dil. acid at room temp. due to their low K_{sp} .

Phosphate (PO_4^{3-})

- 1°, 2° and 3° phosphate salts of alkali metals are water soluble.
- 1° phosphate salts of alkaline earth metals are water soluble whereas their 2° and 3° phosphate salts are water insoluble.
- 3° phosphate salts are water insoluble except alkali metal phosphate salts and $(\text{NH}_4)_3\text{PO}_4$.



Note : AsO_4^{3-} ion also give same colour ppt. with ammonium molybdate soln. but on boiling (distinction between PO_4^{3-} and AsO_4^{3-} ion)

Chromate (CrO_4^{2-}) / Dichromate ($\text{Cr}_2\text{O}_7^{2-}$)

